Atharv Bang

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.42
Fiitjee Junior College	Class XII (Telangana State Board of Intermediate Education)	2024	93.60%
Shree Hanuman Vyayam Shala Public School	Class X (CBSE)	2022	85.60%

WORK EXPERIENCE

DISHA *(Digital Intelligence System for IITH Advancement)* | Intern, AI Developer

- Architected an **offline invoice-processing** pipeline that automatically extracts and validates information from invoices *(PDF, PNG, JPG)*, combining **PyMuPDF for text extraction**, **Tesseract OCR** with **OpenCV preprocessing** as a fallback, and **spaCy NER with regex** for structured field extraction, with results exportable to *JSON or CSV.*
- Implemented **12 rule-based validation checks** that feed into a weighted, **per-field confidence-scoring** system, blending extraction-source reliability with OCR confidence to flag **low-quality extractions.**

AVNIYA CLOUD SOLUTIONS | Intern, ML Engineer

- Fine-tuned a **MobileNetV2** on a custom Indian food image dataset for **food classification**, and integrated **MiDaS depth estimation** with **DeepLabV3 segmentation** to compute food volume and infer portion weight.
- Extended the pipeline with **YOLOv8 for multi-food detection** on a single plate, enabling **per-item weight and nutrient breakdown** with a confidence-scoring system and calibrated weight limits for robustness

PROJECTS

AI Financial Advisor

- Developed an **AI-powered financial advisor** chatbot using a **multi-agent LLM architecture**, with specialized agents for **market analysis**, strategy generation, execution planning, and risk assessment.
- Built an intelligent conversational system that delivers **real-time financial insights** and personalized **investment recommendations**, simplifying complex financial decision-making through coordinated agent workflows.

Finance Analytics & Recommender System

- Designed a stock analytics platform in **Python and Streamlit**, covering **150+ NIFTY-listed stocks** with live market data, **candlestick pattern detection**, and technical indicators feeding a **next-day price movement** classifier.
- Trained **Random Forest and XGBoost models** using a time-based train/test split to avoid data leakage, and built a **K-Means/cosine-similarity recommender system** to surface comparable stocks within an interactive dashboard.

EV Charging Station Optimizer

- Implemented an EV charging station placement system, using **Gonzalez's greedy k-center algorithm** to compute placements on real, directed road networks, guaranteeing a **provable 2-approximation** to the optimal solution.
- Built a custom shortest-path engine using **Dijkstra**, **all-pairs search** alongside a landmark-based approximate distance oracle, cutting **precompute time from 2.8s to 5.7ms** on an **8,100-node city graph.**

SomnoAI | Sleep Quality Prediction from Lifestyle & Academic Patterns

- Developed a **multiclass classification pipeline** in Python to predict student sleep quality from lifestyle and academic data of **500+ students**, benchmarking Logistic Regression, Random Forest, and Gradient Boosting models.
- Achieved **82% accuracy** and **0.95 weighted ROC-AUC** with the best-performing model, **conducted EDA** to identify **screen time as the strongest predictor** of sleep quality **(correlation: 0.63)** via heatmaps and regression plots.

POSITIONS OF RESPONSIBILITY

Sponsorship Head | Abhyanth (Sports Council) Heading the domain, driving **brand partnerships & outreach** to secure event funding. Managing sponsor relations and negotiations to ensure long-term partnership value.

Finance Co-ordinator | Finance & Consulting Club Delivered a hands-on financial markets workshop for students, focusing on technical finance. Assisted in the execution of **Horizon**, gaining direct exposure to large-scale event planning.

Machine Learning Co-ordinator | Tinkerer's Lab Led ML/DL projects addressing campus-specific problems. Coordinated logistics for **THRUST** events, including **UNDER-25 Summit and Candle Night Concert.**

Events Co-ordinator | Inter IIT Sports Meet'25 Coordinated events for **4,500+ participants,** Managed artist logistics and hospitality for performers including **Mohammed Irfan, Pineapple Express and Warsi Brothers.**

RELEVANT COURSES

Technical Courses | Data Structures & Algorithms, Artificial Intelligence, Machine Learning, Programming, R & Python.

Mathematics & Others | Probability & Random Variables, BioStatistics, Linear Algebra, Entrepreneurship, Economics.

TECHNICAL SKILLS

Tools & Programming Languages | C++, Python, SQL, R, VS Code, Jupyter, Google Antigravity, Claude Code, Github.

Core CS Topics | Data Structures and Algorithms, OOPS, Operating System, DBMS & Computer Network Basics.

Machine Learning | Regression & Classification Algorithms, Feature Selection, Evaluation Metrics & Clustering Algorithms

Agentic AI | LangChain, LangGraph, RAG, Vectorless RAG, Deep Agents, GuardRails, LLM Evaluation & LLM Gateways.

Libraries & Frameworks | Numpy, Pandas, Matplotlib, Streamlit, TensorFlow, Keras, PyTorch, OpenCv, XGBoost.

Soft Skills | LeaderShip, Public Speaking, Multi-Team Collaboration, Event Management, Time Management.

SCHOLASTIC ACHIEVEMENTS

- Delivered a comprehensive **2-day AI/ML workshop to 200+ students**, covering key Machine Learning algorithms in depth and guiding them through a **hands-on Recommender Systems** project from concept to implementation.
- Solved **450+ questions** across platforms like Leetcode, GeekForGeeks & Coding Ninjas.
- Won the **Champion Trophy** at the State Championship of **ABACUS**, secured **2nd Runner-up** in the following edition.